Certified Adaptive Examination of Peripheral Blood Smears: When Information-Gain Stopping Fails, and What Can Be Guaranteed Instead

Qudrat Ullah and Balázs Harangi

***Abstract*—Automated examination of peripheral blood smears increasingly decides which fields to inspect and when to stop. We asked what it can guarantee about unexamined material. Using pre-registered simulations from 28,280 expert-labeled cells, two external cell sources, and annotated bone-marrow regions from 99 patients, we evaluated a reimplemented information-saturation stopping rule, derived the examination needed to certify that rare evidence is absent, and tested design-based, anytime-valid certification that the blast proportion exceeds 10% or 20%. The saturation rule's stopping point barely changed with rarity; with one critical field among 200 ten-cell fields, it missed 76% of positive specimens, against 85% for random sampling of equal size. Without location information, any strategy must examine 95% of the fields to certify absence at a 5% error level. Valid tests kept false certification within the nominal level under acquisition shifts that cut blast sensitivity from 0.91 to 0.63 and 0.34, but efficiency did not transfer: classifier-driven sampling needed 0.45 to 0.75 times the verified cells of uniform sampling in distribution, and 1.3 to 3.0 times under shift at margins of 0.05 to 0.10. Near the threshold, a 200-cell count was wrong in up to 27% of specimens, whereas valid procedures were mostly undetermined. Analyzing the cells of sampled fields as independent draws exceeded the nominal error in 27 of 75 real specimens, and single-reader verification certified that reader's count. For automated hematology, certifying a blast-count decision depends on the sampling design, the unit of analysis, and verification, not on classifier confidence.**

***Index Terms*—Adaptive sampling, anytime-valid inference, blast count, blood smear, risk-limiting audit, sequential testing.**

I. Introduction

A peripheral blood smear contains far more cells than any examiner inspects. A manual differential counts 100 or 200 leukocytes, and the variability this implies has long been known [1]. Automated systems add a degree of freedom: they choose which fields to examine and when to stop. For questions that hinge on rare cells, such as the blast proportion relative to a classification threshold [2], [3], what matters is not classifier accuracy on the examined fields but what can be said about the fields that were not examined.

Stopping rules based on *information saturation* answer this implicitly, stopping when further fields would add little information about the selector's representation of the specimen. InfoDPP-PAC [4] is a carefully constructed recent instance for whole-slide histopathology; its authors state that its certificate bounds residual information in the selection objective, not diagnostic performance, and that distinction is our starting point. We ask four questions. RQ1: does information-saturation stopping fail systematically when decision-changing evidence is rare? RQ2: why does certifying the *absence* of rare evidence require near-exhaustive examination? RQ3: how many cells must be examined to certify a threshold decision under adaptive sampling? RQ4: where do the assumptions of anytime-valid audit methods break on real data?

Our contributions are:

- a pre-registered evaluation showing that, in the tested settings, the stopping point of saturation stopping barely changed with rarity, that its halting signal decoupled from critical evidence as cells were pooled into fields, and that on real marrow tiles it did worse than uniform sampling in low-blast specimens (RQ1);
- a minimax lower bound showing that near-exhaustive examination is required of *every* strategy without location information, its consequence for halting rules that ignore verified labels, and a proposition that ties a relevance-ranked search to the model's receiver operating characteristic (ROC) curve (RQ2);
- an evaluation of design-based, anytime-valid certification of blast-proportion assertions [5], [6]: validity, cost from 500 to 4,000 cells against the best valid uniform design, and decisions at fixed budgets (RQ3);
- an empirical examination of the assumptions behind these guarantees (RQ4) under acquisition shift on two external sources, spatial clustering, unequal field sizes and reader disagreement measured on real marrow annotations.

This is a methodological study: the expert who verifies sampled cells is simulated with the datasets' expert labels, most specimens are composites of real single-cell images, and no result establishes clinical diagnostic validity.

II. Related Work

Rümke characterized the binomial and finite-population variability of manual differentials [1]; for absence of rare events, the "rule of three" bounds a rate after zero events [7], and Proposition 1 is its design-based finite-population counterpart. InfoDPP-PAC [4], a recent preprint, selects patches greedily under a blended information-gain and relevance objective and stops when the largest information gain among unselected patches falls below a concentration-corrected threshold. Its certificate rests on a stability assumption whose constant and target are assumed rather than derived; our results do not depend on it. Blast recognition itself is well studied [8]; our question concerns examination and inference.

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Risk-limiting audits bound the probability of confirming a false assertion about a finite population [9]. ALPHA [5] turns assertions into bounded-mean tests through an *assorter*, with a test supermartingale for sampling without replacement [10]–[12]; Shekhar et al. [6] extend this to weighted, adaptive sampling with side information, and stratified extensions exist [13]. Prediction-powered inference [14], prediction-powered e-values [15] and active inference [16] combine predictions with a verified subset, mostly as fixed-sample statements about a superpopulation; our tests are anytime-valid statements about one finite specimen. Two failures we measure are textbook survey sampling, the design effect of cluster sampling [17] and the bias of the mean-of-ratios estimator [18], and classifier-driven sampling is probability-proportional-to-size sampling [19], [20]; their size on real smear arrangements, and their effect on anytime-valid certificates, are not known.

III. Setting and Theory

A. Estimands

A specimen has N fields; field i holds m_i cells, B_i of them blasts, $M = \sum_i m_i$, and $p = \sum_i B_i/M$. We distinguish *detection* (has a critical field, one with at least one blast, been examined?), *estimation* of p, and *threshold certification* ($p > \theta$?). We use $\theta = 0.10$, the blast percentage at which the International Consensus Classification (2022) separates myelodysplastic syndrome with excess blasts from the category MDS/AML [3], and $\theta = 0.20$, the World Health Organization (WHO) 5th-edition threshold for acute myeloid leukemia without defining genetic abnormalities [2]. Both count blast equivalents, which our labels do not; the thresholds are analyzed separately and are inputs to a procedure, not validated here. An examiner selects fields sequentially, possibly using features x and past labels, and a selected field is *verified*: its cells' labels are revealed. All probabilities are over the examiner's randomization for a fixed specimen, so the guarantees are finite-population and design-based. They concern the sampling frame, the fields the imaging system can present; which cells an analyzer images and how the smear relates to the patient's blood are outside it.

B. Certifying absence (RQ2)

Let C be the set of k critical fields and M_n the event that none is among n examined fields; let $h(N,k,n) = \binom{N-k}{n}/\binom{N}{n}$.

Proposition 1 (Uniform sampling): Under uniform sampling of n fields without replacement, $\mathbb{P}(M_n \mid C) = h(N,k,n)$; for $k = 1$ this is $1 - n/N$, so $\mathbb{P}(M_n) \le \alpha$ if and only if $n \ge \lceil (1-\alpha)N \rceil$. For $0 \le n \le N$, $[\max\{0, 1 - n/(N-k+1)\}]^k \le h \le (1 - n/N)^k$, so $n \ge N(1-\alpha^{1/k})$ suffices.

Assumption 1 (No location information): The class of specimens regarded as possible, given the observed features x, contains every placement C with $|C| = k$.

Theorem 1 (Minimax lower bound): Under Assumption 1, every policy, adaptive or not, randomized or not, that examines at most n fields satisfies $\max_{|C|=k} \mathbb{P}_\pi(M_n \mid x, C) \ge h(N,k,n)$. Uniform sampling attains the bound; every deterministic policy has worst-case miss probability 1 if $n \le N-k$.

Proof. Fix x and a realization u of the examiner's randomness. On the run in which every examined field is negative, every input to the policy is determined by (x,u), so it examines a fixed set $A(u)$, $|A(u)| \le n$, and the actual run misses C if and only if $C \cap A(u) = \emptyset$; hence $\mathbb{P}_\pi(M_n \mid x, C) = \mathbb{P}_U(C \cap A(U) = \emptyset)$. Let C be uniform over k-subsets, independent of U, which Assumption 1 permits. For fixed u the fraction of k-subsets disjoint from $A(u)$ is at least $\binom{N-n}{k}/\binom{N}{k} = h(N,k,n)$; averaging over U, and noting that the maximum over C is at least the average, gives the bound (Yao's principle [21]). Uniform sampling attains it by Proposition 1; a deterministic policy examines one set A with $|A| \le N-k$, and placing C outside it gives miss probability 1. □

We verified all statements exactly in rational arithmetic and, for $N \le 7$, by solving the search game as a linear program. If k is unknown, the worst case is $k = 1$, so certifying absence at level α needs $\lceil (1-\alpha)N \rceil$ fields (Fig. 1). The theorem also bears on RQ1: a rule whose halting depends only on the features and its own randomization, not on verified labels, examines a number of fields fixed before any evidence is seen, so with one critical field its worst-case miss probability is at least $1 - \bar{n}/N$ when it examines $\bar{n}$ fields on average. If verification misses an examined critical field with probability β, absence cannot be certified below β even by a census.

Informative features can let a search beat the bound on average, through a model of where evidence lies:

Proposition 2 (Ranked search): Let one field be critical with score $S_c \sim F_1$, and the other $N-1$ scores be independent with distribution F_0. A scan of the n highest-scoring fields misses the critical field with probability $\mathbb{E}_{S_c}[\mathbb{P}(\mathrm{Bin}(N-1, 1 - F_0(S_c)) \ge n)]$, which tends to $1 - \mathrm{TPR}(t_r)$ as $N \to \infty$ with $n/(N-1) \to r$, where t_r is the threshold with false-positive rate r, TPR is the true-positive rate, and F_0 is continuous and strictly increasing at t_r.

Given $S_c = s$, the critical field is missed if and only if at least n others score above s; the limit follows by dominated convergence. With $F_1 = F_0$ the formula returns $1 - n/N$. Certifying absence by ranking thus needs a sensitivity of about $1-\alpha$ at a false-positive rate of n/N, a property of the model that no sampling design supplies. It is an average-case result under independence, neither a proof that only ranking helps nor a worst-case guarantee: a ranked scan surely misses a critical field placed among low-scoring fields.

C. Certifying a threshold assertion (RQ3)

Making "$p > \theta$" true requires more than θM blasts, which cannot hide in one field, so the bound above does not apply. With equal field sizes and the assorter $a_j = b_j/(2\theta)$, b_j being the blast fraction of the j-th sampled field, $\bar{a} > 1/2 \iff p > \theta$, a one-sided bounded-mean assertion. Under uniform sampling of fields, ALPHA [5] gives a test supermartingale; rejecting when it exceeds $1/\alpha$ controls false certification at α at any data-dependent stopping time. For adaptive sampling, write $p = \sum_i w_i f_i$ with known weights and $f_i \in [0,1]$. If the field I_t is drawn from a predictable distribution q_t, the observation $Z_t = w_{I_t} f_{I_t}/q_t(I_t)$ has conditional mean at most the unsampled mass (equal if q_t is positive on every

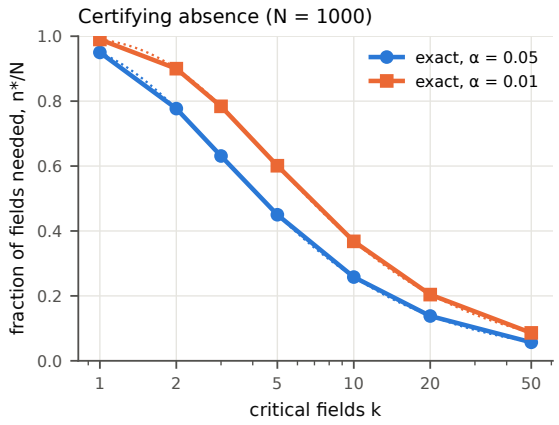


Fig. 1. Fraction of fields that any strategy without location information must examine to certify the absence of k critical fields ($N = 1000$; exact values solid, sufficient closed form $1 - \alpha^{1/k}$ dotted).

unsampled field), which under $H_0 : p \leq \theta$ is at most $\mu_t = \theta - \sum_{s<t} w_{I_s} f_{I_s}$; a betting process $\prod_s (1 + \lambda_s (Z_s - \mu_s))$ with predictable bets $0 \leq \lambda_s \leq 1/\mu_s$ is a test supermartingale whatever the classifier does [6]. We use their prop-MS sampler, $q_t(i) \propto w_i S_i$ with a floor (S_i the field's mean classifier score), with approximate-Kelly bets and, exploratorily, hedged bets (the average of the processes with $\lambda_s = \kappa/\mu_s$, $\kappa \in \{0.05, 0.1, 0.2, \ldots, 0.9\}$). Validity requires only known q_t and verified labels; positivity of q_t is needed for power and for "$p < \theta$" certificates, not for validity of "$p > \theta$", because $f_i \geq 0$.

However, the average of the field proportions, $N^{-1} \sum_i B_i/m_i$, is not p when field sizes differ.

Proposition 3 (Unequal field sizes): (a) If every m_i is known, $w_i = m_i/M$ and $f_i = B_i/m_i$ give $\sum_i w_i f_i = p$, so the weighted test is valid for $H_0 : p \leq \theta$. (b) If only $m_i \leq m_{\max}$ is known, $a_i = (B_i - \theta m_i + \theta m_{\max})/m_{\max} \in [0, 1]$ and $N^{-1} \sum_i a_i > \theta \iff p > \theta$, so equal-weight tests on a_i with null mean θ are valid.

For (b), $0 \leq B_i \leq m_i \leq m_{\max}$ bounds a_i, and $\sum_i a_i = (\sum_i B_i - \theta M)/m_{\max} + N\theta$. The margin shrinks by $\bar{m}/m_{\max}$, so a loose bound costs power.

Remark 1 (Imperfect verification): Suppose the verifier counts the same M cells as the reference and, in this specimen, its false blast calls number at most β_0 times the true non-blasts. Then $\tilde{p} \leq p + \beta_0(1 - p)$, so testing $\tilde{p} > \theta + \beta_0(1 - \theta)$ is valid for $p > \theta$; symmetrically, testing $\tilde{p} < \theta(1 - \beta_1)$ is valid for $p < \theta$ if missed blasts number at most β_1 times the true blasts. These are per-specimen bounds that no design supplies; average disagreement rates are not such bounds, and excluding cells the reference counts can raise $\tilde{p}$ without any false blast call.

We compare, on the same specimen, ALPHA and betting with uniform sampling; betting with classifier-driven sampling, with and without a control variate; an invalid *uncorrected* design that ranks fields by score but analyzes them as uniform draws; and conventional 100- and 200-cell counts. Uniform ALPHA, the best valid design without a classifier in our runs, is the reference for cost ratios.

IV. Experimental Design

A. Pre-registration and status of analyses

Hypotheses, decision rules, seeds and scripts were frozen in five git tags, each pushed before the corresponding data were generated; first confirmatory outputs followed the push by 43 s to 18 min (audit table in the repository). "Confirmatory" means fresh Monte Carlo draws on a disjoint seed family over the same cell pool, which protects against analytic flexibility but not against effects specific to that pool; only H4a–c and H7a used data not analyzed before. All later analyses, including a rerun of all cost comparisons on the same specimens for the three cell sources (seed family 20261101), used new seed families and are labeled exploratory. The four pre-registered sensitivity analyses were run after the main analyses, with the frozen script and their pre-registered seeds; all deviations are logged in the repository.

B. Data

KU-Optofil [22]: Sysmex DI-60 images of May–Grünwald–Giemsa peripheral blood smears labeled by hematologists; we used 28,280 cells from 275 identified patients (13 classes) in five patient-grouped folds, excluding 3,204 cells without patient identifiers. Blasts come from only 11 patients (489 blasts). *RV-PBS* [23]: 752 annotated cells (153 blasts) from phone-camera fields, used only as an acquisition-shift probe. *MLL23* [24]: 41,621 peripheral-blood cells (Pappenheim stain, Metafer scanner), 8,606 myeloblasts; we excluded atypical promyelocytes (blast equivalents our labels do not count), smudge cells and normoblasts, leaving 36,529; there are no patient identifiers. *BoneMarrowWSI* [25], [26]: pediatric bone-marrow aspirate slides with 228 exhaustively annotated regions of about 225 μm; we used the annotations, the per-round reader labels, and the image pixels of the annotated regions. All four sources are public and de-identified; no new data were collected from human participants.

C. Representation, teachers and specimens

Cells were embedded with a frozen DINOv2 ViT-S/14 [27]; a logistic probe cross-fitted by patient fold reached an out-of-fold area under the ROC curve (AUC) of 0.979. The pre-registered *cell-level teacher* scores a field by its largest cell score, which presupposes cell detection; an exploratory *field-level teacher* is a logistic regression on the pooled field embedding, the representation the selector sees. Part I composites have $N = 200$ fields of $m \in \{1, 5, 10, 20\}$ cells (mean-pooled embeddings) with $k \in \{1, 2, 5, 10, 20\}$ critical fields of one blast. Part III composites have 100 fields of 5 cells ($M = 500$) with p fixed exactly, blasts drawn from all 489 blast images and background cells from blast-free patients; 10% of that background are platelets and erythroblasts, so p is the blast share of imaged objects (a leukocyte-only background changed nothing material). Exploratory extensions use M up to 4,000 and field sizes drawn from BoneMarrowWSI.

D. InfoDPP-PAC and baselines

No code has been released. We implemented the objective, greedy blended gain and halting rule (Eqs. 3–7 of [4]) with the published weights, set $\rho = 1$, and calibrated τ on blast-free specimens of other folds so that the median stop is 16.3% of all fields. The original runs stopped at 16.3% of a 300-patch budget, about 2.4% of a slide (a ratio of reported averages, not the original configuration), so our setting gives InfoDPP-PAC about seven times its own examination share; we also report a 2.4% run. A random seed of 5 fields fits the Gaussian process and is not counted as examined. We used DINOv2 instead of the paper's Phikon encoder, which was trained on hematoxylin and eosin histology, because there is no established open encoder for May–Grünwald–Giemsa-stained blood cells. Baselines were uniform sampling at the same per-episode cost (paired), a fixed field order, confidence-based stopping (stop at the first field with relevance ≥ 0.9 or after 30 consecutive fields ≤ 0.1; illustrative thresholds, not matched in cost), and exhaustive examination.

E. Statistics

Miss-rate differences are reported with t intervals on the five patient-grouped folds: episodes resample cells from one pool of 275 patients (blasts from 11), so the pre-registered exact McNemar tests on paired episodes measure Monte Carlo precision only, and a bootstrap over five folds has only 126 distinct resamples. False certification: Clopper–Pearson intervals and one-sided exact binomial tests of rate ≤ 0.05. Costs: paired Wilcoxon tests with Holm correction and ratios of medians with bootstrap intervals; real-data analyses bootstrap over patients; $\alpha = 0.05$ unless stated. Validity claims rest on design-based theorems that hold for every specimen; miss rates and costs are estimates for this pool. Our ALPHA and betting statistics agree with the reference implementation (SHANGRLA) to within 3×10^{-13} in log scale on 2,000 random cases, apart from a conservative rule missing from our frozen ALPHA that changed no median cost; a suite of 50 deterministic tests, with adversarial samplers and negative controls, is in the repository.

V. Results

A. Information-saturation stopping (RQ1)

In 24,000 confirmatory episodes the median stop was 32–33 of 200 fields in every cell, varying by at most one field across rarity levels (H1a). This is an empirical finding: the halting statistic has no relevance term but depends on the embeddings and the selected set. The AUC with which the information-gain term separates critical from non-critical fields fell from 0.778 at $m = 1$ to 0.593, 0.552 and 0.525 at $m = 5, 10, 20$, whereas the field-level teacher on the same pooled embeddings kept 0.978, 0.925, 0.882 and 0.827 (Kill-test 1; the pre-registered cell-level teacher gave 0.977 at $m = 1$ to 0.896 at $m = 20$). The evidence survives pooling; the halting signal does not use it.

At $k = 1$, InfoDPP-PAC missed the critical field in 0.315, 0.681, 0.759 and 0.805 of specimens at $m = 1, 5, 10, 20$; uniform sampling of the same size missed 0.822, 0.830, 0.846 and 0.828, matching $1 - n/N$ (H1d; Fig. 2). The advantage shrank with dilution for every k (H1b), but the pre-registered claim of no advantage at $m \geq 10$ failed (H1c): the difference was -0.087 at $m = 10$ (fold-level t interval $[-0.153, -0.021]$; pre-registered episode-level interval $[-0.107, -0.067]$) and -0.023 $[-0.050, 0.003]$ at $m = 20$. Dilution, not rarity, governed the gap; rarity drove the absolute miss rate of every method. With one critical field, the fixed field order at the same cost missed 0.83–0.85, like uniform sampling, as expected for composites without geometry; confidence-based stopping halted after a median of 10–33 fields and missed 0.74–0.90, close to uniform sampling at its own cost (0.77–0.92). Critical-evidence recall, the fraction of the k critical fields examined, fell with dilution for InfoDPP-PAC from 0.34–0.69 at $m = 1$ to 0.19–0.21 at $m = 20$, against 0.15–0.18 for uniform sampling and the fixed order. Because InfoDPP-PAC's halting does not use verified labels, the consequence of Theorem 1 noted in Section III-B applies: at a mean of 32.4 examined fields, its worst-case miss probability with one critical field is at least $1 - 32.4/200 \approx 0.84$, and the observed rates are averages over placements that the features partly reveal. Replacing its teacher by the field-level teacher hardly changed its misses (0.32, 0.66, 0.76 and 0.77; exploratory), because the teacher enters only the selection objective, with weight 0.25, and not the halting statistic.

Two other saturation rules, pure information-gain sampling and k-center coverage stopping, calibrated to the same cost on 4,000 new specimens, missed within 0.04 of InfoDPP-PAC in every cell. A scan ranked by the field-level teacher missed 0.05, 0.14, 0.21 and 0.37, against 0.051, 0.146, 0.221 and 0.394 predicted by Proposition 2: relevance, not the stopping criterion, separated the rules. At about 2.4% of the specimen, InfoDPP-PAC made 4–5 selections after its seed; its advantage was -0.17 $[-0.27, -0.08]$ with single-cell fields and at most about 0.03 otherwise.

On real image tiles (exploratory; BoneMarrowWSI, 3,040 tiles of about $56\,\mu$m from 99 patients, specimens of 16–32 tiles, 200 runs each), InfoDPP-PAC stopped after 5.4 tiles on average. In the 24 specimens with at most a quarter of tiles critical it missed every critical tile in 0.80 of runs against 0.70 for uniform sampling ($+0.105$ $[0.002, 0.203]$; worse in 18 of 24), and in at least 95% of runs in 15 specimens, as Theorem 1 predicts for a nearly deterministic selector. In the same specimens a raster missed 0.67 and a scan ranked by the tile-level teacher 0.54, a teacher whose AUC was only 0.69 there; in the 14 specimens with a single critical tile, k-center stopping, which examined 8.5 tiles on average, missed it in 0.89 of runs against 0.70 for uniform sampling at its cost ($+0.19$ $[0.03, 0.34]$).

B. Certified threshold decisions (RQ3)

In 12,000 confirmatory 500-cell specimens, all four valid designs kept false certification at $p = \theta$ between 0.001 and 0.022 (H3a), whereas the uncorrected design certified falsely in 0.993 ($\theta = 0.10$) and 0.889 ($\theta = 0.20$) of boundary specimens (H3b; Fig. 3). Classifier-driven betting needed

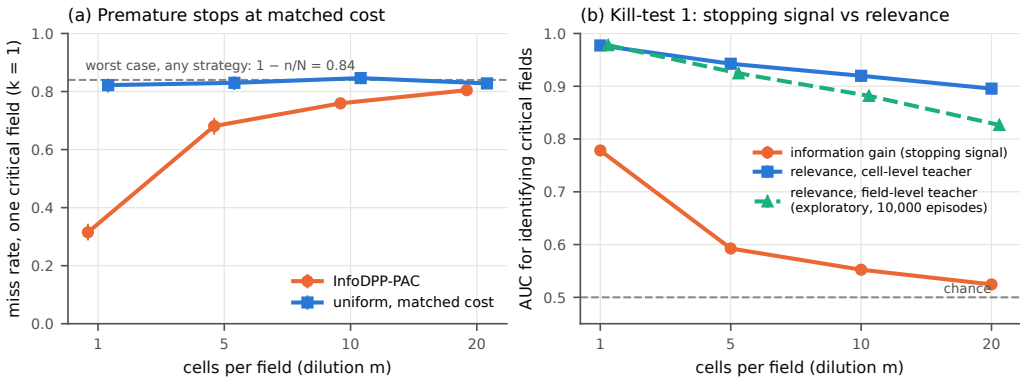


Fig. 2. RQ1, confirmatory run. (a) Miss rate with one critical field for InfoDPP-PAC and for uniform sampling of the same size (95% Wilson intervals); dashed: the worst-case bound $1 - n/N$. (b) AUC of the information-gain term and of the cell-level teacher for ranking critical above non-critical fields; dashed, exploratory: the field-level teacher on the same pooled embeddings as the selector (10,000 separate episodes).

fewer cells than uniform ALPHA at every $p > \theta$ (H3c), 0.45–0.75 times as many, for example 75 vs. 165 at $p = 0.15$, $\theta = 0.10$, and cost fell with the margin (H3d). Uniform betting with approximate-Kelly bets needed 1.03–1.83 times the cells of ALPHA, so savings measured against it (0.28–0.73) are overstated; with hedged bets (exploratory, 4,400 specimens) it matched ALPHA (0.98–1.17), while classifier-driven medians changed by at most 14%.

With specimens of up to 4,000 cells (exploratory; 17,600 specimens), false certification stayed ≤ 0.040 at $\alpha = 0.05$ and ≤ 0.007 at $\alpha = 0.01$, and cost was governed mainly by the margin (Table I): at $p = 0.15$ against $\theta = 0.10$, uniform ALPHA needed a median of 165, 205 and 245 cells at $M = 500$, 1,000 and 4,000 and classifier-driven betting 80, 80 and 85. At margins of 0.05 or more and $M = 4{,}000$, uniform ALPHA was within 20% of Wald's approximation $\log(1/\alpha)/\mathrm{KL}(p\|\theta)$ [28]; within 0.02 of the threshold it still grew with M, to 1,583 ($\theta = 0.10$) and 2,033 cells ($\theta = 0.20$) at $M = 4{,}000$, so the near-census at $M = 500$ was a small-specimen effect. Classifier-driven sampling saved 36–65% of the cells of uniform ALPHA at every M, except at $p = 0.22$ against $\theta = 0.20$, where the saving was 19–23% at $M \leq 1{,}000$ and at most 3% for $M \geq 2{,}000$. Savings count expert-verified cells; every field must still be imaged and scored. Lowering α to 0.01 multiplied the cost by 1.0–1.7. Design-based tests need no calibration set, unlike conformal risk control [29], whose finite-sample correction leaves no risk budget at $\alpha = 0.01$ below 99 calibration episodes.

At fixed budgets (exploratory; 2,000-cell specimens, cells sampled individually), we compared three-way decisions ("above", "below", "undetermined"; Table II). Within 0.02 of the threshold a 200-cell count, the usual peripheral-blood differential, was wrong in 11–27% of specimens (100 cells: 18–37%). Valid procedures, two one-sided sequential ALPHA tests or exact hypergeometric tests at $\alpha/2$ each, were wrong in at most 0.7% of specimens off the threshold, but at those margins returned "undetermined" in 61–98% of specimens at budgets up to 500 cells. With a budget fixed in advance and uniform sampling, a fixed-n test is decisive more often; anytime validity pays when the examiner stops on the data or samples adaptively. The sequential tests used prior guesses of $\theta \pm 0.05$, fixed in advance; with ALPHA's default guess, which suits field-level observations, they certified "above" in only 31% instead of 86% of specimens at $p = 0.15$, $\theta = 0.10$ and $n = 500$. Conversely, for fields, guesses of $\theta + 0.05$ and $\theta + 0.10$ kept ALPHA valid but made it up to 2.2 times slower than the default (exploratory).

C. Where the assumptions break (RQ4)

Classifier error. On KU-Optofil, blast recall at 95% specificity varied across the 11 blast patients from 0.39 to 1.00, false-positive rates across blast-free patients were overdispersed 6.6-fold, and by class were highest in immature granulocytes (0.26–0.33), reactive lymphocytes and basophils (0.19–0.20); within cell class, errors were associated with the color, brightness, contrast and sharpness of the image. Certification stayed valid with blasts drawn only from low-recall patients (≤ 0.007), at 2.3 times the cost.

Acquisition shift. Applied unchanged, the KU-trained probe's blast sensitivity fell to 0.340 [0.250, 0.421] on RV-PBS (H4a) and 0.63 [0.62, 0.64] on MLL23 (AUC 0.786 and 0.915), while its false-positive rate barely rose (0.058 and 0.060 against 0.050; H4b not supported). Every design stayed valid (H4c; all ≤ 0.034), but efficiency did not transfer (Fig. 4(a)). Against uniform ALPHA on the same specimens (exploratory rerun), classifier-driven sampling needed 1.30 [1.18, 1.37] and 1.46 [1.36, 1.75] times the cells on MLL23 at $p = 0.15$ and 0.20, and 1.67 [1.53, 1.75] and 3.00 [2.79, 3.33] times on RV-PBS; the confirmatory comparison with uniform betting gave 1.77 on RV-PBS at $p = 0.20$ (H6d). A 50% uniform mixture stayed valid (H6a); with approximate-Kelly bets it kept 58–73% of the saving over uniform betting (H6c) but cut the RV-PBS excess only to $+23\%$ [11, 30], missing the pre-registered 20% bound (H6b), and a 25% uniform share gave essentially no protection (ratio 1.73); with hedged bets it needed 0.61–0.81 times the cells of ALPHA in distribution, 0.77–0.82 on MLL23 and 0.95–1.15 on RV-PBS (exploratory).

Spatial clustering and the unit of analysis. Analyzing the cells of sampled fields as independent draws ignores the design effect [17]: in synthetic arrangements (18,000 specimens),

TABLE I

MEDIAN CELLS TO CERTIFY $p > \theta$ AGAINST SPECIMEN SIZE M (EXPLORATORY; $\alpha = 0.05$; 400 SPECIMENS PER CELL; 5 CELLS PER FIELD). U: ALPHA WITH UNIFORM SAMPLING; C: BETTING WITH CLASSIFIER-DRIVEN SAMPLING; APPROX.: $\log(1/\alpha)/\mathrm{KL}(p\|\theta)$. †BLAST IMAGES REUSED, WHICH AFFECTS ONLY C. THE $M = 500$ COLUMN IS A SEPARATE RUN FROM THE CONFIRMATORY ONE IN FIG. 3, SO ITS MEDIANS DIFFER BY UP TO 10 CELLS.

		$M = 500$		$M = 1{,}000$		$M = 2{,}000$		$M = 4{,}000$		
θ	p	U	C	U	C	U	C	U	C	Approx.
0.10	0.12	390	220	655	420	1,040	660	1,583	798	1,425
0.10	0.15	165	80	205	80	220	85	245†	85†	245
0.10	0.20	65	40	68	40	65	40	75†	40†	67
0.20	0.22	435	335	810	660	1,338	1,300	2,033†	2,043†	2,455
0.20	0.25	250	145	303	175	338†	190†	420†	185†	406
0.20	0.30	100	60	115	65	120†	65†	125†	70†	106

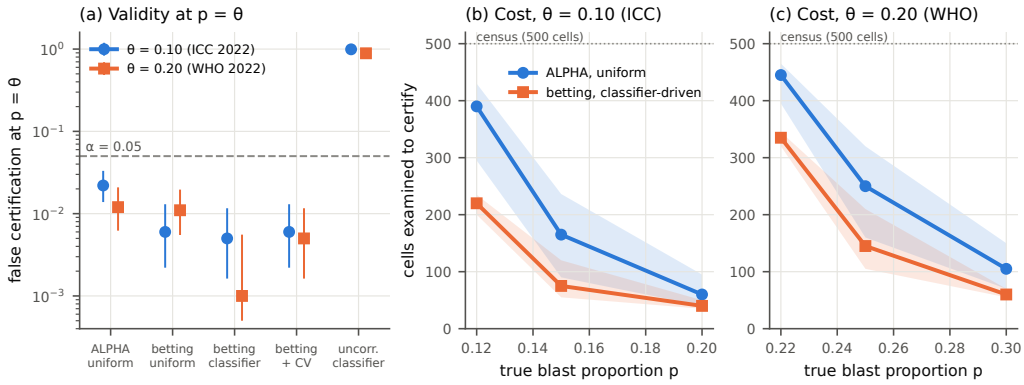


Fig. 3. RQ3, confirmatory run (500-cell specimens; $\alpha = 0.05$). (a) False certification at $p = \theta$ with 95% Clopper–Pearson intervals (log scale); CV: control variate; the uncorrected design analyzes classifier-ranked fields as uniform draws. (b, c) Median cells to certification with interquartile bands; specimens that never certify count as 500 cells.

TABLE II

DECISIONS AFTER n INDIVIDUALLY SAMPLED CELLS (EXPLORATORY; $M = 2{,}000$; 2,000 SPECIMENS PER ROW). CONV.: SHARE WRONG FOR THE CONVENTIONAL n-CELL COUNT; EXACT (FIXED-n HYPERGEOMETRIC) AND SEQ. (SEQUENTIAL): SHARE CORRECT / WRONG FOR VALID TESTS AT $\alpha/2$ PER SIDE; THE REST IS "UNDETERMINED"; ϵ: < 0.01.

		$n = 200$			$n = 500$		
θ	p	Conv.	Exact	Seq.	Conv.	Exact	Seq.
0.10	0.08	0.11	0.10/0	0.06/ϵ	0.03	0.39/0	0.15/ϵ
0.10	0.12	0.22	0.15/ϵ	0.05/ϵ	0.06	0.34/0	0.15/ϵ
0.10	0.15	0.01	0.60/0	0.31/ϵ	0.00	0.96/0	0.86/ϵ
0.20	0.18	0.19	0.12/ϵ	0.04/ϵ	0.08	0.23/0	0.09/ϵ
0.20	0.22	0.27	0.09/ϵ	0.03/ϵ	0.12	0.25/ϵ	0.08/ϵ
0.20	0.25	0.05	0.38/ϵ	0.16/0	0.00	0.87/0	0.59/0

under the strongest clustering it certified falsely in 0.327 ($\theta = 0.10$) and 0.478 ($\theta = 0.20$) of boundary specimens (H5b), whereas field-level designs stayed valid (≤ 0.042, H5a), although clustering raised the cost of uniform field sampling 1.6 to 2.2-fold (H5c). In 116 exhaustively annotated marrow regions of 75 patients, blasts were clustered (H7a; permutation $p = 0.024$ at the quadrant and 0.0005 at the 4×4 scale) but weakly: the bias-corrected intraclass correlation was 0.013 [0.002, 0.028] and 0.043 [0.018, 0.067] (exploratory), a design effect of about 1.2 for 5-cell fields, and a null-centered estimate of 0.059 [0.007, 0.108] at 4×4 in the 32 regions with at most 30% blasts. We then used each patient's regions as a real specimen, with 4×4 subregions as fields and its own proportion as threshold (an exact null; 400 sampling orders per specimen; exploratory). The weighted test of Proposition 3(a) certified falsely in 0.001 of runs and at most 0.01 in any specimen; the cell-level analysis gave 0.051 overall, yet exceeded 0.05 in 27 of 75 specimens (21 significantly; up to 0.335), including 7 of the 23 with at most 20% blasts. Validity is a statement about each specimen, and a pooled rate can hide failures. These specimens are small (median 105 cells) and mostly high-blast: only two had blast proportions between 0.05 and 0.30, so arrangements near the diagnostic thresholds are barely represented, and our synthetic stress arrangements were one to two orders of magnitude more clustered than these monolayer regions.

Unequal field sizes. Marrow quadrants hold 12–29 nucleated cells (interquartile range). With these sizes, averaging field proportions, the mean-of-ratios estimator [18], certified falsely in 0.38 [0.35, 0.41] and 0.32 [0.29, 0.35] of boundary specimens even with blasts placed independently of size, and in every specimen with blasts enriched in small fields: the test is valid for the mean of proportions, which is the wrong quantity. In the real regions this mean differed from the blast proportion by only -0.004 on average $[-0.008, 0.001]$, but at a boundary specimen even a small difference decides the outcome. Both constructions of Proposition 3 stayed at or below 0.005; with known counts, classifier-driven sampling certified $p = 0.15$ against $\theta = 0.10$ with 297 cells against 646 for uniform sampling, and the bounded assorter needed 631

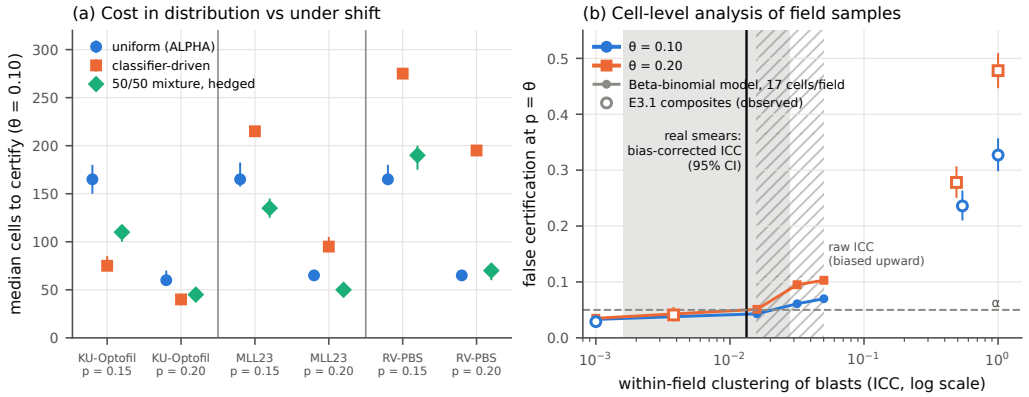


Fig. 4. RQ4. (a) Median cells to certify $p > 0.10$ with 95% bootstrap intervals, in distribution (KU-Optofil) and under two acquisition shifts (MLL23, RV-PBS): uniform ALPHA, classifier-driven betting and a 50% mixture with hedged bets on the same specimens (exploratory rerun, 400 specimens per point); all designs kept false certification ≤ 0.034. (b) False certification at $p = \theta$ when cells of sampled fields are analyzed as individual draws: composites with scattered or clustered arrangements (markers) and a Beta-binomial model with 17 cells per field (lines). Solid band and line: bias-corrected clustering of real marrow regions (exploratory); hatched: the raw, upward-biased estimate, for comparison.

because $\bar{m}/m_{\max} \approx 0.24$.

Verifier error. BoneMarrowWSI publishes each reading round's labels, not who read which cell [25]. The second reader called a blast in 1.1% [0.7, 1.7] of the first reader's counted non-blasts and a non-blast in 1.2% [0.8, 1.6] of its blasts; 2.7% of cells were counted by one reader only. With the first reader as verifier and the consensus as truth (83 specimens), the reader's proportion differed from the consensus by a median of 0.005 (maximum 0.075). Each specimen was tested at its own consensus proportion and examined almost completely, so the sign of the reader's error decided the outcome: "$p > \theta_s$" was certified falsely in all 23 specimens in which the reader over-counted (0.28 of runs), "$p < \theta_s$" in all 52 in which it under-counted (0.63), and in at most 0.008 of runs in the 8 without error. The test certified the reader's count, as designed. Remark 1 with average disagreement rates, which are not per-specimen bounds, did not repair this (0.16, 0.42); a margin of 0.03, selected on the same specimens, sufficed for "above" but not for "below".

One benchmark. Table III puts these designs and conditions side by side: where their assumptions held, design-based tests kept false certification at or below 0.034, whereas every analysis without a design-based guarantee failed in some condition, even where it looked acceptable in distribution (cells as draws, 0.041) or pooled over real arrangements (0.051). Validity transferred across conditions; efficiency did not.

D. Robustness

The pre-registered Part I sensitivity analyses changed no direction (difference from uniform sampling at $k = 1$, fold-level t intervals). With max pooling it was -0.115 $[-0.198, -0.032]$ at $m = 10$ and -0.056 $[-0.141, 0.029]$ at $m = 20$, and InfoDPP-PAC still missed 73% and 77% of specimens; with single-cell representations it was -0.036 $[-0.079, 0.007]$ at $m = 10$; and across operating points of 5–50% of the fields it never exceeded 0.13 in absolute value at $m \geq 10$. A single-host background enlarged the residual advantage (-0.27 $[-0.38, -0.16]$ at $m = 10$ and -0.15 $[-0.19, -0.10]$ at $m = 20$), although InfoDPP-PAC still missed 57% and 71% of specimens; that construction draws up to 4,000 background cells with replacement from about 110 cells of one patient, so "the advantage vanishes by $m = 20$" depends on the background. Varying the four InfoDPP-PAC settings that the original paper does not report (ρ, kernel length-scale, noise variance and seed size; exploratory, 2,000 new specimens) changed its miss rate and its difference from uniform sampling by at most 0.03, with no significant change (smallest Holm-adjusted McNemar $p = 0.38$). Of the 15 pre-registered empirical hypotheses 12 were supported, and the 5 implementation checks passed (Table IV); the proposal's rarity hypothesis (Kill-test 2) had already failed on exploratory data and was replaced by H1b before the confirmatory run.

VI. Discussion

Our results separate three quantities that are often reported together. *Model confidence* about examined fields says nothing about unexamined ones. *Information saturation* concerns the selector's representation; its association with decision-relevant evidence fell toward chance as more cells were pooled into each field, although the pooled representation still carried the evidence. *Design-based certification* bounds the probability of a false threshold decision for the specific specimen, given known selection probabilities and verified labels. Only the third concerns what was not examined; it held in every condition we tested, specimen by specimen, while the classifier changed how many cells it needed, in either direction. It concerns the imaged frame and the verifier's labels, not the patient. Validity needed known selection probabilities, analysis at the sampled unit, the cell-weighted target and verified labels (Tables III and V); an accurate classifier with exchangeable errors was needed only for efficiency.

Five practical consequences follow. First, certify a proportion against a stated threshold rather than the absence of rare cells, which requires a near-census; its cost in verified cells is governed mainly by the margin. Second, record selection probabilities, analyze at the unit that was sampled, and weight

TABLE III

ONE BENCHMARK ACROSS CONDITIONS (STORED RESULTS; NO NEW SIMULATION). TOP: SHARE OF SPECIMENS CERTIFYING "$p > \theta$" AT $p = \theta$ ($\alpha = 0.05$; LARGER OF $\theta = 0.10, 0.20$; 400–1,000 SPECIMENS); BONEMARROWWSI: POOLED OVER RUNS WITH $\theta_s = p_s$ [LARGEST SPECIMEN RATE]. BOTTOM: MEDIAN CELLS TO CERTIFY $p = 0.15$ AGAINST $\theta = 0.10$ (UNCERTIFIED SPECIMENS COUNT ALL CELLS). SPECIMENS: 500 CELLS; ABOUT 1,300 WITH UNEQUAL FIELDS; MEDIAN 105 REAL CELLS. STRESSED: ASSUMPTION TESTED (UNIT: ANALYZED UNIT = SAMPLED UNIT; TARGET: CELL-WEIGHTED p). C/X: CONFIRMATORY/EXPLORATORY RUN. [a]UNEQUAL FIELDS AND BONEMARROWWSI: ALPHA ON THE BOUNDED ASSORTER, BETTING WITH KNOWN COUNTS (PROP. 3). [b]COINCIDES WITH ALPHA FOR EQUAL FIELDS; 1.00 WITH BLASTS ENRICHED IN SMALL FIELDS. [c]EXPLORATORY RERUN, ALL DESIGNS ON THE SAME SPECIMENS; UNIFORM ALPHA SEES ONLY LABELS. ONE READER WITH MARGIN 0.03: 0.0003 [0.013].

	Composites of KU-Optofil cells				External cells		BoneMarrowWSI	
	In distr.	Clustered	Confusable backgr.	Unequal fields	RV-PBS	MLL23	Real arrangem.	One reader
Stressed	—	*unit*	*classifier*	*target*	*classifier*	*transfer*	*geometry*	*verifier*
Status	C	C	X	X	C	X	X	X
False certification of "$p > \theta$" at $p = \theta$								
ALPHA, uniform[a]	0.022	0.025	0.031	0	0.026[c]	0.034[c]	0 [0]	—
Betting, uniform[a]	0.011	0.005	0.011	0.005	0.011	0.011	0.001 [0.010]	0.277 [1.00]
Betting, classifier-driven	0.005	0.003	0.007	0.001	0.001	0.001	—	—
Betting, 50% mixture	0.014	—	—	—	0.003	0.004	—	—
Hedged betting, 50% mixture[c]	0.028	—	—	—	0.025	0.030	—	—
Ranked, analyzed as uniform	0.993	—	—	—	—	—	—	—
Cells of fields as draws	0.041	0.478	—	—	—	—	0.051 [0.335]	—
Mean of field proportions[b]	—	—	—	0.379	—	—	0.062 [1.00]	—
100-cell count, "above"	0.427	0.388	—	—	—	—	—	—
Median cells to certify $p = 0.15$ against $\theta = 0.10$								
ALPHA, uniform[a]	165	345	160	1,031	165[c]	165[c]	—	—
Betting, uniform[a]	270	355	265	646	260	283	—	—
Betting, classifier-driven	75	55	105	297	265	215	—	—
Betting, 50% mixture	145	—	—	—	285	220	—	—
Hedged betting, 50% mixture[c]	110	—	—	—	190	135	—	—

TABLE IV

PRE-REGISTERED HYPOTHESES (E: EMPIRICAL CLAIM; V: IMPLEMENTATION OR THEOREM CHECK). KT2, THE PROPOSAL'S RARITY HYPOTHESIS, WAS TESTED ON EXPLORATORY DATA ONLY AND IS NOT AMONG THE 20.

ID		Hypothesis (abridged)	Verdict
KT2	E	Gap widens as evidence becomes rarer	**no**
H1a–b	E	Stop point unchanged across rarity; advantage shrinks with dilution	yes
H1c	E	No advantage (± 0.05) at $m \geq 10$	**no**
H1d	V	Uniform miss rate $= 1 - n/N$	yes
KT1	E	Information gain and relevance separable	yes
H3a	V	Four designs valid at both thresholds	yes
H3b–d	E	Uncorrected selection invalid; classifier-driven cheaper; cost falls with margin	yes
H4a	E	Blast recall drops under shift	yes
H4b	E	False-positive rate rises under shift	**no**
H4c, H5a, H6a	V	Valid with shifted scores, under clustering, with a mixture	yes
H5b–c	E	Cell-level analysis invalid; clustering raises cost	yes
H6b	E	Mixture caps loss under shift below 20%	**no**
H6c–d	E	Mixture keeps $\geq$50% saving; reversal replicates	yes
H7a	E	Blasts clustered in real smears	yes

fields by their cell counts. Third, keep a uniform component in classifier-driven sampling, and judge savings against the best valid uniform design. Fourth, allow and report "undetermined": near the threshold it is the honest outcome of a few hundred cells, and a 200-cell count that forces a call was wrong in up to 27% of specimens. Fifth, a certificate concerns the verifier's count; relating it to the true proportion needs a bound on verifier error, established on separate data or, in future work, from a random subsample of double-read fields. A procedure that is valid but usually undetermined near the threshold is statistically sound and operationally limited; making it decisive requires more cells, not a more confident model.

Limitations. Most specimens are composites from one laboratory, and all KU-Optofil blasts come from 11 patients, so cost and miss-rate comparisons are imprecise. The real-data spatial, tile and verifier analyses use small, mostly high-blast pediatric bone-marrow specimens, not peripheral blood, tested against each specimen's own proportion rather than diagnostic thresholds; they test the method, not clinical use. The external sources lack patient identifiers, so their intervals understate uncertainty; the reading rounds are not identified experts; and in all other analyses verification is simulated with the datasets' labels. Several analyses, including every cost comparison against uniform ALPHA on external cells and all hedged-bet results, are exploratory, and the pre-registration tags are timestamped by the hosting service only. InfoDPP-PAC is a reimplementation calibrated to a cost, though two other saturation rules gave the same result. We used one frozen encoder, costs count verified cells, not imaging or computing time, and we did not implement prediction-powered e-values [15].

TABLE V
WHAT EACH GUARANTEE ASSUMES, AND WHAT HAPPENED WHEN THE ASSUMPTION WAS TESTED.

Assumption	Needed for	Test and result
No location information	Theorem 1	Features informative on composites: ranking helps on average, through a model (Proposition 2), not for every placement
Selection probabilities known	Validity	Uncorrected ranking: false certification 0.89–0.99
Analysis unit equals sampled unit	Validity	Cells as draws (design effect ignored): up to 0.34 per real specimen, 0.33–0.48 under strong clustering
Target is the cell-weighted p	Validity	Averaging proportions: 0.32–0.38; Proposition 3 restores ≤ 0.005
Verified label equals truth	Validity for the true p	Single reader: false certification wherever its count lay on the claimed side (0.28 "above", 0.63 "below")
Accurate classifier, exchangeable errors	Efficiency only	Validity held under shift (≤ 0.034); cost relative to uniform ALPHA rose from 0.45–0.75 to 1.3–3.0 at margins of 0.05–0.10

VII. CONCLUSION

Information saturation does not certify what remains unexamined, and without location information certifying the absence of rare evidence requires examining nearly the whole specimen. Threshold decisions about the blast proportion can instead be certified with finite-sample, anytime-valid guarantees. A classifier can reduce the cells an expert must verify in distribution and increase them under shift, so it should be combined with a uniform component; validity rests on the sampling design, the unit of analysis and weighting, and the reliability of verification. Fully annotated peripheral-blood smears with patient identifiers, and verification by identified hematologists, are the most important next steps.

DATA AND CODE AVAILABILITY

Code, the pre-registration with its git history and audit table, the deviation log and the analysis outputs are available at https://github.com/Afridi-7/certified-adaptive-pbs (to be archived on Zenodo with a DOI). KU-Optofil, BoneMarrowWSI-PediatricLeukemia and MLL23 are public under CC BY 4.0; RV-PBS is distributed by its authors without a stated license and was used for research only.

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